ElimuSentry - Phase 2: Building the Preprocessing Pipeline

What was built, what the diagnostics showed, and where reality changed the plan

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04-652 Artificial Intelligence System Design, Assignment 2, Phase 2 (Strategy Execution, Pipeline and Visual Diagnostics)
Repository: https://github.com/Emile-Lucky-Muhigira/AISD_Project

***Abstract*—Phase 1 planned, with evidence, how every piece of information about a child in the Ghana Socioeconomic Panel Survey would be prepared for a school-dropout early-warning model. This report covers Phase 2: turning that plan into working code and checking it. The plan was built as one scikit-learn `Pipeline` around a `ColumnTransformer`, with custom preparation steps in `src/transformers.py`. It is fitted on the model-building children only (X_train, 1,307 children from 2009/10) and then applied unchanged to the testing children (X_test, 1,395 children from 2013/14). The output has exactly the planned size: 33 features become 30, and then 53 numeric columns for XGBoost or 48 for logistic regression, with no missing value left. The three removed features can be rebuilt exactly from the kept ones (0 differences in 3,921 checks), so the structural removals add no extra loss beyond the retained source fields. Eight diagnostic figures confirm each step. Most decisions were carried out as planned; five points changed or were sharpened by what the data showed. A retained fixed CPI adjustment aligns nominal monetary units; other cross-period changes remain descriptive. MI-only feature selection is unstable across folds; a fixed structural/domain rule determines the removals. The hours-missed rule is compared descriptively with median filling; no equivalence claim is made. The "was missing" flags are switched on about half as often in the later period. And rank correlation was needed because the ordinary correlation would have missed one repeating pair.**

***Index Terms*—data preprocessing, scikit-learn pipeline, leakage prevention, visual diagnostics, school dropout, Ghana panel survey**

I. Introduction

Aim of this phase. Phase 1 used the Ghana Socioeconomic Panel Survey [2] and decided how to fill gaps, treat extreme values, remove repeating features, encode categories, rescale numbers and split the data with fold-fitted preprocessing [1]. Phase 2 has three tasks, which follow the assignment: (1) implement that strategy as fully runnable code with a strict X_train / X_test separation; (2) check every assumption with eight diagnostic figures; and (3) record where reality agreed with the plan and where it forced a change.

What is in the repository. `src/transformers.py` holds the custom preparation steps; `src/pipeline_builder.py` assembles them into the pipeline; `phase2_implementation/notebooks/pipeline_execution.ipynb` runs everything from start to finish, saves the eight figures to `figures/` at 300 DPI, writes the processed tables to `data/processed/` and the decision log to `phase2_implementation/decision_log.csv`. The notebook was run from top to bottom without errors.

Scope. Phase 2 is about preparing the data. A quick five-fold run of two models is included only to prove that the full pipeline (preparation plus model) works end to end with fold-fitted preprocessing. Tuning the models, adjusting the risk percentages (calibration) and the single final test belong to the modelling phase, so testing outcomes are not used for fitting, tuning, calibration or threshold selection. Aggregate temporal outcomes were inspected descriptively, so confirmatory independence is not claimed.

Reading this report. Technical terms are explained in plain language in the Appendix. Section II describes what was built, Section III the figures, Section IV the decision log, Section V the changes and their reasons, and Section VI the checks on the output.

II. What Was Built

One pipeline, two versions. The pipeline is built with scikit-learn [4]. The same steps produce two versions of the data (Table I). The **XGBoost version** [5] keeps numbers in their own units, because a decision tree only compares values; it has 53 columns. The **logistic regression and PCA version** adds the log on long-tailed features and standardises every number, and leaves out one option per label as the reference; it has 48 columns. This follows Phase 1, C1 exactly.

How leakage is prevented. Every step that learns something (averages, middle values, neighbours, cap limits, money ranks, the list of options, the scaler) is inside the pipeline. The pipeline is *fitted* on X_train only and then only *applied* to X_test. Inside cross-validation it is fitted again on the building part of each fold. The notebook checks this directly: it saves the fitted pipeline, applies it to X_test, and confirms that nothing it learned has changed [6]. The only step applied outside the pipeline, the inflation adjustment, uses a fixed published number, not the data.

The split. Inputs contain 1,307 training and 1,395 later-period children. Primary temporal evaluation is restricted to the 731 children in 605 new households (76 dropouts), with no shared household or person. Secondary evaluation uses 1,175 new people but allows 370 shared households. Five

TABLE I
THE PIPELINE, STEP BY STEP, IN THE ORDER THE DATA PASS THROUGH IT

#	Step (code name)	What it does	Learns from	Phase 1
0	`deflate_to_2009_prices`	Divides 2013/14 money amounts by 1.62 to express them in 2009/10 prices	Nothing (fixed number)	C1
1	`MissedHoursRule`	A blank "hours missed" becomes 0 when attendance was recorded	Nothing (fixed rule)	B1
2	`MissingFlagger`	Adds the four yes/no "was missing" flags before any gap is filled	Nothing (fixed rule)	B1
3	`GradeKNNImputer`	Mixed-distance grade KNN; k selected in inner district-grouped training folds; recalculates over-age	X_train (donors, numeric ranges and inner-selected k)	B1
4	`ColumnDropper`	Removes people per room, lost a parent and school grade (33 → 30)	Nothing (fixed list)	B4b
5	`CategoryRecoder`	Turns 1/2 answers into 0/1, reverses the textbooks scale, groups relationship to head into 3 options	Nothing (fixed rule)	C1
6	`SimpleImputer` (in the `ColumnTransformer`)	Fills gaps: average for balanced numbers, middle value for lopsided ones, most common answer for yes/no and ordered items, a "missing" option for school type	X_train (averages, middle values, most common answers)	B1
7	`LogIQRCapper`	Caps travel time, household size and rooms at the IQR limit on the log scale; adds the log for the simple model	X_train (limits)	B2, B3
8	`TrainingRank`	Turns the two money amounts into a 0–1 rank against the model-building children	X_train (reference values)	C1
9	`OneHotEncoder`	One yes/no column per option of each label	X_train (list of options)	C1
10	`StandardScaler` (simple model only)	Puts every number on a common scale (average 0, spread 1)	X_train (average, spread)	C1

Steps 1–5 run one after another on the whole table because each one needs the result of the one before (for example, flags must be added before gaps are filled). Steps 6–10 run side by side on different groups of columns inside the `ColumnTransformer`. Step 0 is applied to X_test before the pipeline. Code: `src/transformers.py` and `src/pipeline_builder.py`.

training folds keep whole districts together. Aggregate later-period covariates and outcomes were already explored, so this is retrospective temporal evaluation, not an untouched external confirmation. These subsets are fixed by IDs, not their outcomes.

The inflation factor. Phase 1 promised to convert 2013/14 money to 2009/10 prices using Ghana's consumer price index. The interview dates in the raw survey files show that the first visit took place from November 2009 to April 2010 and the second, labelled 2013/14, mostly from March to October 2014, with a few households in 2015. Using World Bank annual consumer price inflation for Ghana [3] (index 2009 = 100: 2010 = 110.7, 2014 = 172.6, 2015 = 202.3), weighted by the number of interviews in each year, the price level was 107.5 at the first visit and 174.0 at the second. Every 2013/14 amount is therefore divided by $174.0/107.5 = 1.62$.

III. VISUAL DIAGNOSTICS

Missing answers (Fig. 1.1). The pipeline sees the same pattern as Phase 1: 3,847 of 43,131 feature cells missing in X_train (8.92%) and 1,827 of 46,035 in X_test (3.97%); 53 complete children in X_train and 467 in X_test. After the pipeline, **no missing value is left** in either table.

Outcome (Fig. 1.2). Dropouts are the small group in both periods, so each dropout is given extra weight during training (`scale_pos_weight` for XGBoost, `class_weight="balanced"` for logistic regression), recalculated inside every fold.

Extreme values (Fig. 2.1). The limits learned by the pipeline (IQR rule on the log scale [8]) are exactly those planned (294.0 minutes, 15.2 people, 18.8 rooms), and exactly 13 values are capped in X_train (2 travel times, 6 household sizes, 5 room counts) and 5 in X_test. The two money features are no longer shown: as planned in Phase 1, C1, they are ranked instead of capped, so their extreme values cannot dominate.

Repeating features (Figs. 3.1 and 3.2). The same two pairs are flagged as in Phase 1: father alive and lost a parent ($\rho = -0.850$), rooms and people per room ($\rho = -0.855$). The third case, school grade, is an exact calculation from age and years over-age (panel c). The fitted lines confirm that each removed feature follows directly from features that are kept.

Links with dropout (Fig. 3.3). The ranking (mutual information [9]) is the one reported in Phase 1: region first (mutual information 0.037), then household goods, rooms, school spending, English score, household size and parents' education. No feature is removed for a weak link: choosing features belongs inside the folds of the modelling phase.

Transformations (Fig. 4.1). The log is the step that changes the shape; Min–Max and standardising only change units. This confirms the Phase 1 choice: log on long tails, then `StandardScaler` for the simple model and PCA, and no rescaling for XGBoost.

Dimension reduction (Fig. 5.1). The first component captures 16.6% of the variation, and the curve then rises slowly: 21 of 48 components are needed for 90% and 28 for 95%. There is no small set of components that summarises the children well, so PCA would save little and would make the features harder to explain, which matters for an early-warning system whose warnings must be explained [12]. PCA is therefore kept as a diagnostic and not used to replace the features [7].

IV. DECISION LOG: PLAN VS IMPLEMENTATION

Table II is generated from the executed decision log. It records the mixed-distance KNN revision, fixed structural removal, explicit assumption checks and stricter temporal cohorts. Calibration and final temporal prediction scoring remain deferred to the modelling phase.

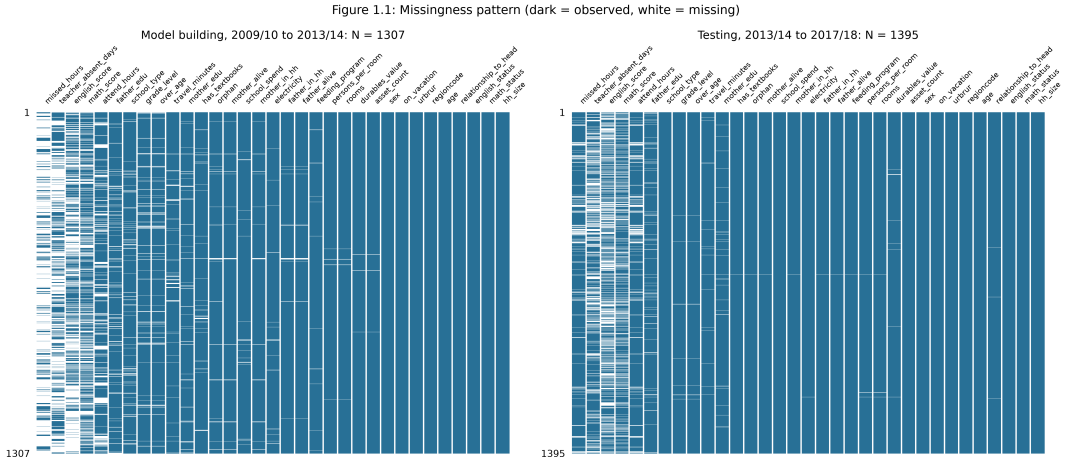


Fig. 1.1. Where answers are missing. Left: X_train; right: X_test. *How to read it:* each thin row is one child and each column one feature; white marks are missing answers. The large white blocks on the left of X_train are hours missed, days teacher absent, the two test scores and hours attended. X_test has far fewer gaps because the 2013/14 tablets required an answer.

TABLE II
DECISION LOG: EVERY PLANNED STEP FROM PHASE 1 AGAINST WHAT WAS BUILT IN PHASE 2

Step	Planned in Phase 1	Implemented in Phase 2 (evidence)	Status
Fix recording errors	Fixed rules (codes, impossible values)	Applied in 01_build_dataset; nothing learned	As planned
Inflation adjustment	Convert 2013/14 money to 2009/10 prices (CPI)	Factor 1.62 (World Bank CPI, weighted by interview months)	Fixed external price adjustment; other distribution changes remain
Split	By time; primary new households; secondary new people	X_train 1307, all temporal 1395, primary 731, secondary 1175	As planned
Hours missed	Blank = 0 when attendance recorded; median otherwise	Rule applied; CV PR-AUC (LR) 0.266 vs 0.275 with median fill (descriptive comparison; uncertainty retained)	Sensitivity comparison; no formal equivalence claim
Was-missing flags	4 flags before filling	teacher absent, hours attended, father's and mother's education	As planned
Was-missing flags across periods	Compare flags between periods	Flags are on about half as often in 2013/14 (tablets) as in 2009/10 (paper)	Confirmed limitation
Filling gaps	Mean / median / mode by skew; 'missing' option for school type	SimpleImputer fitted on X_train	As planned
School grade	Mixed-distance KNN with training-only k selection	Inner-grouped k=15 on full training; over-age recalculated	Revised and validated
Repeating features	Spearman |rho| >= 0.85, VIF > 10; drop 3; re-check in each fold	MI-only diagnostic matched the structural list in 1/5 folds; removal is fixed from source identities/domain meaning, not MI; rebuilt exactly including unknowns	Fixed structural list; MI-only check is descriptive
Extreme values	Log-IQR cap on travel, household size, rooms	Limits 294.0, 15.2, 18.8; 13 capped (train)	As planned
Money	Rank 0-1 against model-building values	TrainingRank fitted on X_train	As planned
Encoding	One-hot, ordinal, 0/1	53 columns (XGBoost), 48 (LR)	As planned
Scaling	None for XGBoost; log + StandardScaler for LR and PCA	PCA: 21 components for 90%, 28 for 95%	As planned
Cross-validation	5 folds, stratified, grouped by district	StratifiedGroupKFold; 0 shared districts or households	As planned
Class imbalance	Class weighting, recalculated per fold	scale_pos_weight / class_weight='balanced'	As planned
Calibration, final test	Training-only nested tuning/calibration; frozen retrospective temporal evaluation	Not run in Phase 2 (modelling phase)	Deferred

The same table is saved by the notebook as `phase2_implementation/decision_log.csv`.

V. DEVIATIONS AND RATIONALE

A. Price adjustment and rank interpretation

Retain the fixed external CPI conversion (factor 1.62) and learn the rank reference on training rows only. Training-reference ranks bound long-tailed monetary inputs without fitting the later distribution. Previously explored later-period ranks differ from the training median, but that alone cannot identify a causal price or wealth change: composition, measurement and attrition can contribute. No claim is made that inflation explains most of the change or that school policy

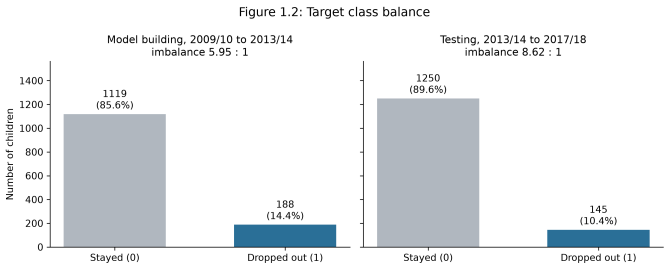


Fig. 1.2. Children who stayed in school and who dropped out. X_train: 188 of 1,307 dropped out (14.4%, imbalance 5.95 : 1); X_test: 145 of 1,395 (10.4%, 8.62 : 1).

TABLE III
RE-CHECK OF THE REPEATING FEATURES INSIDE EACH FOLD

Fold	ρ rooms / people	ρ father / lost parent	Less info, pair 1	Less info, pair 2	Least info, group 3
1	-0.851	-0.842	people/room	lost parent	grade
2	-0.861	-0.831	rooms	lost parent	grade
3	-0.854	-0.861	rooms	lost parent	grade
4	-0.858	-0.854	people/room	father alive	age
5	-0.850	-0.866	rooms	lost parent	age

Computed on the building part of each fold only. Group 3 = age, grade and years over-age. Full table: section 7 of the notebook.

TABLE IV
SIZE OF THE DATA AT EACH STAGE

Stage	X_train	X_test
Children (rows)	1,307	1,395
Features before the pipeline	33	33
Features after removing 3 repeating ones	30	30
Columns, XGBoost version	53	53
Columns, logistic regression / PCA version	48	48
Missing values after the pipeline	0	0
Children deleted	0	0

53 = 16 numbers or ordered numbers + 9 yes/no items + 24 yes/no columns from the label options + 4 "was missing" flags. The logistic regression version leaves out one option for each of the 5 labels (53 − 5 = 48).

TABLE V
NO INFORMATION LOST: REBUILDING THE REMOVED FEATURES

Removed	Rebuilt from kept features	Children checked	Children that differ
School grade	age − years over-age − 5	1,307	0
Lost a parent	Three-valued logic from parent-alive items; unknown remains unknown	1,307	0
People per room	household size / rooms	1,307	0

All training children, including equal missing values; 3,921 checks in total.

caused the spending difference. Within-wave test ranking is not used to fit the primary procedure or to choose it from test scores.

B. Which repeating feature to drop is not stable across folds

The removal list is fixed from feature construction and domain meaning, not from full-training dropout associations. Retain household size and rooms instead of derived crowding; retain parent-alive source fields instead of derived orphan status; retain age and the domain-relevant over-age signal instead of grade. The separate MI-only diagnostic agrees in 1 of 5 folds (Table III), so it is not represented as a stable selection algorithm. This diagnostic does not change the fixed list. Source identities are checked on all training records, including matching missing values; this is a check of structural removal, not a claim that all preprocessing is lossless.

C. Hours missed: an assumption with a sensitivity comparison

The retained rule assumes zero missed hours when attendance is recorded; it is not recovery of a known survey answer. It sets 756 blanks to zero and leaves 261 holiday/unknown-attendance cases for the later median fill. The untuned five-fold comparison gives logistic-regression PR-AUC 0.266 for the rule versus 0.275 for median filling, and XGBoost 0.252 versus 0.257. These are descriptive scores with substantial fold variation; no formal equivalence or no-harm conclusion is supported. The sensitivity is retained so a later training-only nested modelling comparison can assess the choice. Temporal test scores must not select it.

D. Missing indicators can change meaning across periods

Flags are on less often in the later cohort: teacher absent 56.8% vs 31.8%, hours attended 21.8% vs 11.7%, father's education 7.7% vs 5.4%, mother's education 3.8% vs 1.9%. Survey mode, coding and cohort composition may all matter; tablet use alone is not established as the cause. Retain the four design/family-context flags as a fixed modelling choice, not because an unadjusted significance test selected them. Compare their behaviour descriptively across periods and retain the uncertainty in transport claims.

E. Rank correlation was needed, not the ordinary one

The assignment's screening rule is written as $|r| \geq 0.85$. To capture monotone curved relationships and reduce sensitivity to extreme values, Phase 1 used the rank (Spearman) correlation ρ with the same threshold. Phase 2 shows why this mattered: for rooms and people per room, $\rho = -0.855$ but the ordinary (Pearson) correlation is only $r = -0.563$, because the relationship is curved (people per room = household size / rooms). With Pearson r, this repeating pair would have been missed, and the VIF check also misses it (VIF 1.77). For father alive and lost a parent, both measures give -0.850.

VI. PIPELINE OUTPUT VERIFICATION

Final size. Table IV confirms that the pipeline produces exactly the size planned in Phase 1: 33 features become 30, then 53 columns for XGBoost and 48 for logistic regression, for every child of X_train and X_test, with no missing value and no child deleted.

Feature count reduced without information loss. The assignment asks for proof that the reduction from 33 to 30

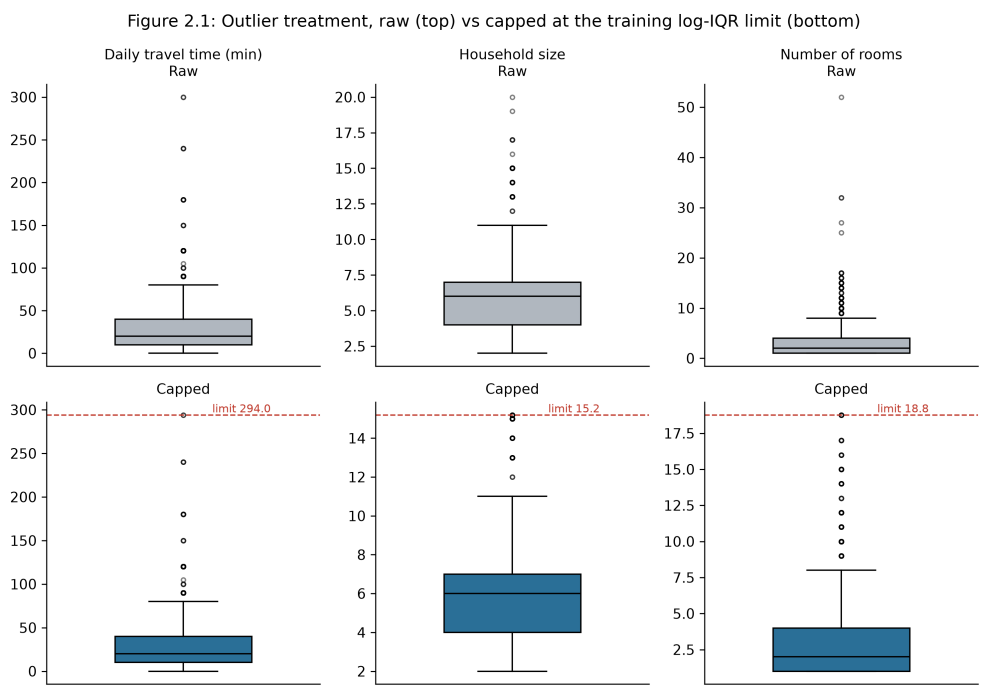


Fig. 2.1. Extreme values before (top) and after (bottom) capping, for the three capped features. *How to read it:* the box holds the middle half of the children and the line inside it is the middle value; dots are unusually high values. The red dashed line is the limit learned from X_train: 294.0 minutes, 15.2 people and 18.8 rooms. Values above the line are pulled down to it; no child is deleted.

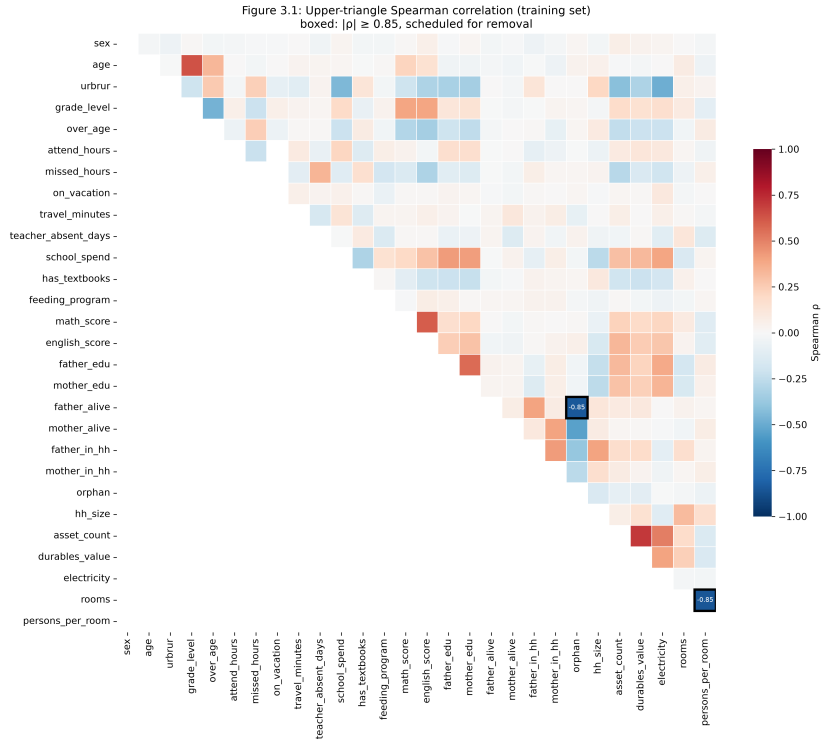


Fig. 3.1. Which features repeat each other (rank correlation, X_train). *How to read it:* each square compares two features; dark red means they rise together, dark blue that one falls as the other rises. The two boxed squares are the pairs at or above 0.85 (ignoring the sign), which are scheduled for removal.

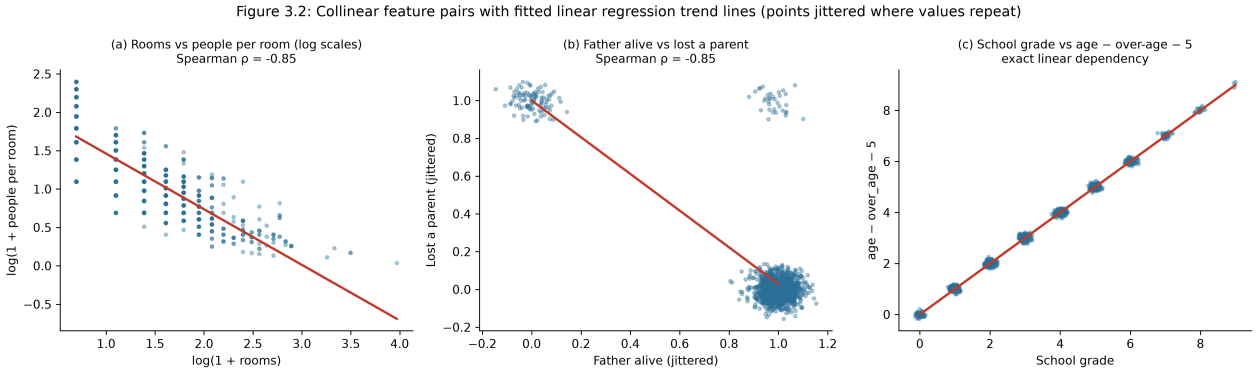


Fig. 3.2. The three repeating features, one dot per child, with a fitted straight line. (a) More rooms, fewer people per room (log scales). (b) "Lost a parent" is almost the opposite of "father alive" (dots spread slightly so they do not sit on top of each other). (c) School grade equals age − years over-age − 5 for every child, so every dot lies on the line.

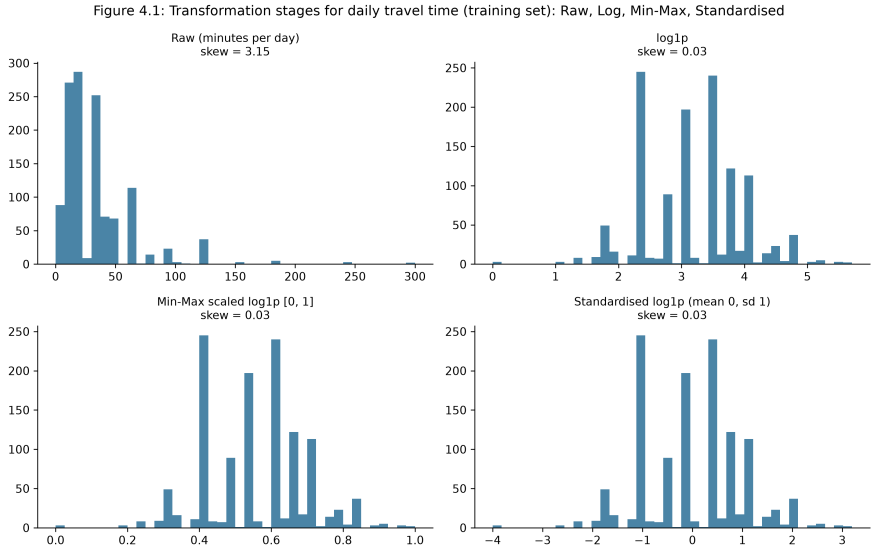


Fig. 4.1. What the transformation stages do to daily travel time (X_train). Top left: raw minutes, with a long tail to the right (skew 3.15). Top right: after the log, the shape is balanced (skew 0.03). Bottom: the same logged values squeezed into 0–1 (Min–Max) and standardised (average 0, spread 1); the shape does not change, only the units on the bottom axis.

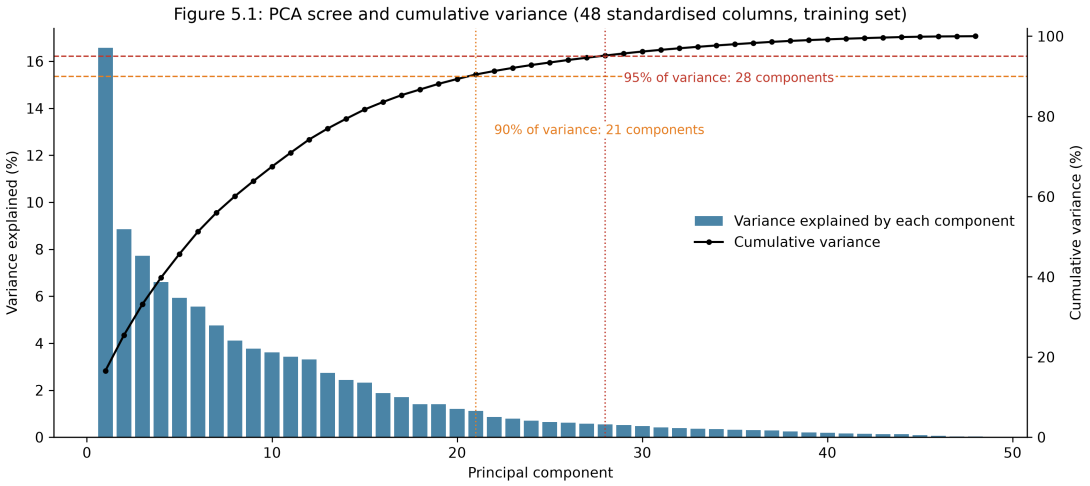


Fig. 5.1. Principal component analysis of the 48 standardised columns of X_train. *How to read it:* each bar is the share of the total variation captured by one combined component; the black line adds them up. The dashed lines mark 90% and 95% of the variation, reached with 21 and 28 components.

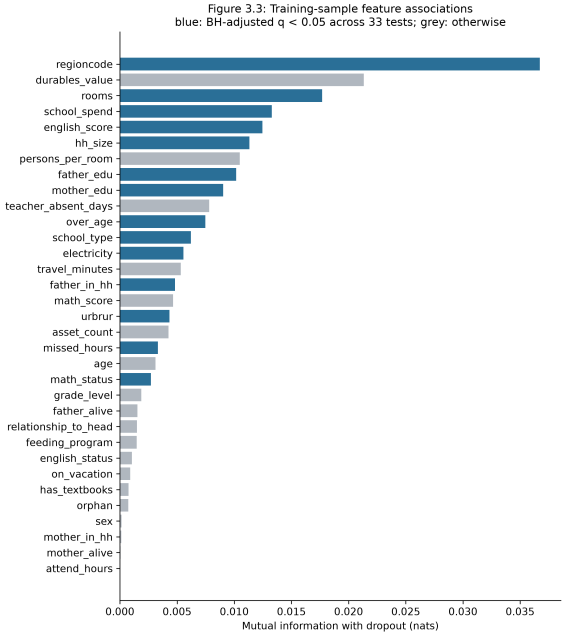


Fig. 3.3. Which features tell us most about dropout (mutual information, X_train). *How to read it:* the longer the bar, the more the feature helps to tell dropouts from children who stay. Blue: BH-adjusted $q < 0.05$ across 33 training-sample tests; grey: otherwise. Sparse tables use Fisher methods; these are exploratory associations.

TABLE VI
QUICK END-TO-END CHECK (5 DISTRICT-GROUPED FOLDS, X_TRAIN)

Model (untuned)	PR-AUC	ROC-AUC
Logistic regression (48 columns)	0.266 (±0.072)	0.627
XGBoost (53 columns)	0.252 (±0.089)	0.602
No skill (dropout rate)	0.144	0.500

Mean (± spread) over the 5 checking parts. Both models are untuned; this checks execution and exploratory training discrimination, not confirmed transport performance.

loses no information. Table V rebuilds each removed feature from features that are kept and compares the result with the original value for every child. The rebuilt value is identical in all 3,921 checks. This establishes the structural identities before subsequent value transformations. Imputation, capping and ranking still intentionally change information; the full pipeline is not claimed to be invertible.

No leakage. (1) Every learned setting is fitted on X_train only; the notebook confirms that applying the pipeline to X_test leaves its fitted settings unchanged. (2) In cross-validation the whole pipeline is re-fitted on the building part of each fold; no district or household is shared between building and checking parts. (3) Later outcomes are not used for fitting or selection. Prior descriptive inspection is disclosed; primary households and people are disjoint.

The pipeline works end to end. Table VI shows that both untuned models, trained through the full pipeline fold by fold, show exploratory training discrimination above the no-skill reference (PR-AUC [11], about 0.25–0.27 against 0.144 for no skill). The numbers are modest, as expected before tuning and with folds made of whole districts, which stops a model from relying on local patterns it has already seen. The processed tables are saved as `data/processed/train_processed_xgb.csv` and `test_processed_xgb.csv` (53 columns plus identifiers and the outcome) and the matching `_lr` files (48 columns); the test files include metadata flags for repeated people, the primary new-household cohort and the secondary new-person cohort. These flags are not model features.

VII. LIMITATIONS AND NEXT STEPS

- **Change over time.** Dropout fell from 14.4% to 10.4%, regional patterns changed, household goods rose in real terms and school spending fell. A model built on 2009/10 may overstate risk in the later period; this will be measured on the single final test.
- **Flags across periods.** The "was missing" flags are switched on about half as often in 2013/14 (Section V-D); their weight in the model will be compared between periods.
- **Children who could not be found again.** As reported in Phase 1, they share known dropout risk factors, so the true dropout rate is probably understated; the sensitivity check on the identity rule (age change of 1–8 years) is part of the modelling phase.
- **Next steps (modelling phase).** Tune both models inside the same folds, adjust the risk percentages (calibration) on the checking parts, choose the cut-off for flagging the riskiest 20%, and then freeze the procedure, then report primary new-household and secondary new-person temporal metrics. Seek a new external/future cohort for confirmation.

APPENDIX: PLAIN-LANGUAGE GUIDE TO THE TECHNICAL TERMS

The terms below are used in this report; terms already explained in the Phase 1 report keep the same meaning.

Pipeline: A chain of preparation steps followed by a model, run in a fixed order. Fitting it learns every setting from the training data at once.

ColumnTransformer: The part of the pipeline that applies the right preparation to each group of columns side by side (for example, a middle-value fill for lopsided numbers and one-hot encoding for labels).

Fit / transform: *Fit* = learn the settings (e.g. a middle value or a cap limit) from data; *transform* = apply the stored settings to any data without learning anything new.

X_train / X_test: The children used to build the model (2009/10) and the children kept aside to test it (2013/14).

Leakage: When information from the data used for checking sneaks into building the model, making results look better than they really are.

Fold, building part, checking part: Cross-validation splits X_train into 5 parts; the model is built on 4 (the building part) and checked on the 5th (the checking part), rotating 5 times.

Custom transformer: A preparation step written for this project (in `src/transformers.py`) that follows the same fit / transform rules as the standard scikit-learn steps.

KNN (nearest neighbours) filling: Filling a missing grade from similar training children with numerical differences and category matches treated separately; neighbour count selected within training.

Capping: Pulling an extreme value down to a set limit instead of deleting the child.

Rank (0–1): The position of a value among the training values: 0 = lowest, 0.5 = middle, 1 = highest.

Consumer price index, inflation factor: The official measure of how prices rise; dividing 2013/14 amounts by 1.62 expresses them in 2009/10 prices.

One-hot encoding: Turning a label with several options (e.g. 10 regions) into separate yes/no columns.

Standardising: Putting numbers on a common scale with average 0 and spread 1.

Correlation (r, ρ): How closely two features move together, from -1 to $+1$. The ordinary (Pearson) r only sees straight-line relationships; the rank (Spearman) ρ also sees curved ones.

VIF: How much a feature is already explained by all the others together (above 10 = redundant).

Mutual information: How much knowing a feature reduces uncertainty about dropout; it captures any kind of relationship.

PCA, component, scree plot: PCA combines many columns into a few new ones (components) that keep as much of the variation as possible. A scree plot shows how much variation each component keeps, and the cumulative line how much the first k components keep together.

Class weighting: Counting each dropout several times more during training so the model pays enough attention to the rare group.

PR-AUC, ROC-AUC: Scores from 0 to 1 for how well a model puts real dropouts at the top of its risk list. For PR-AUC a model with no skill scores the dropout rate (0.144); for ROC-AUC it scores 0.5.

Calibration: Adjusting risk scores so that "20% risk" really means about 1 in 5 such children drop out.

Tie-break rule: The Phase 1 rule used when two repeating features carry similar information: keep the one measured directly and remove the one calculated from others.

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